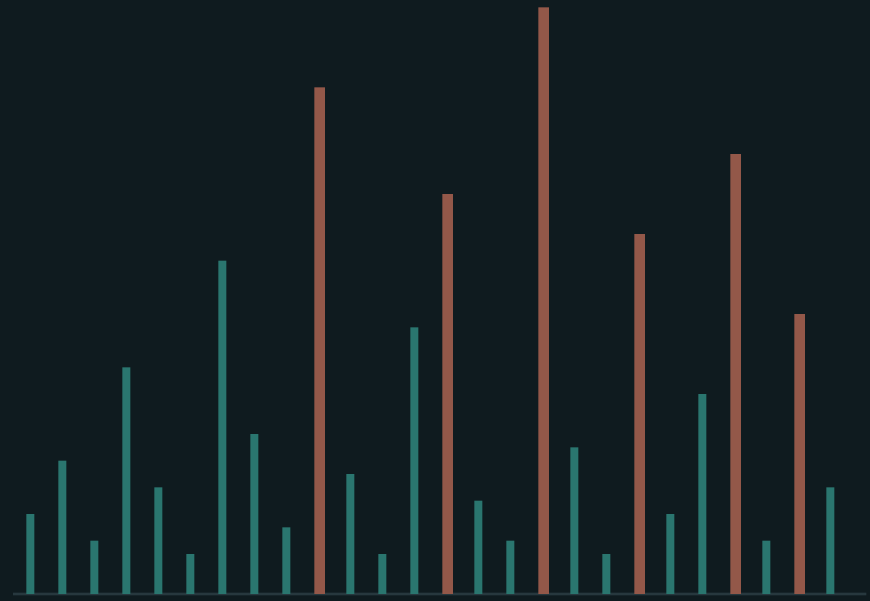
AI HEALTHCARE HACKATHON · OCTOBER 2026

SmartScalpel

An AI that reads surgical smoke to tell tumour from healthy tissue, while the surgeon is still cutting.

[Team names] · University of Leeds



CONTEXT · THE STORY

One in five women go back to theatre

18.5%

of 55,297 women who had breast-conserving surgery in England needed a second operation within 3 months.

IN THEATRE

The surgeon removes the lump and a rim of healthy tissue, judging the edge by sight and touch.

DAYS LATER

The lab finds cancer cells at the edge of the removed tissue: a positive margin.

BACK TO THEATRE

Another excision, or a mastectomy. 7.7% of the women in the study ended up losing the breast.

Source: Jeevan et al., BMJ 2012 · English NHS trusts, 2005–2008

BIOLOGY OF TUMOURS

Tumours rebuild the tissue’s chemistry

Adipose (fat)

Fat cells packed with stored triglycerides. Most of the breast’s volume.

Fibrous stroma

Collagen-rich support tissue. Right next to a tumour it changes too: tumour-border stroma.

Tumour

Dense, fast-dividing cells that build new membranes, which shifts their phospholipid and fatty-acid make-up.

Burn each one and the smoke carries a different **lipid fingerprint**. Tumour can look and feel like the dense stroma around it, but its chemistry gives it away.

TYPE OF SURGERY

Margins are only checked after surgery

1. **Find the tumour**
   Imaging, then a wire or marker shows the surgeon where it is.
2. **Cut with diathermy**
   An electrosurgical knife cuts and seals vessels. It turns tissue into smoke, which is sucked away and thrown out.
3. **Take a rim of healthy tissue**
   The aim is a clear edge with no cancer cells, while keeping as much breast as possible.
4. **Wait for pathology**
   The verdict on the margin comes days later, from the lab.

18.5%

needed a reoperation within 3 months (55,297 women, England)

+£2,136

extra direct cost per patient who needs one (median reoperation: £4,511)

Sources: Jeevan et al., BMJ 2012 · Grant et al., British Journal of Surgery 2019

DATASET · THE HEALTH ASPECT

Real tissue, labelled by pathologists

72	689	3	8,000
patients	smoke spectra	tissue classes	m/z channels each

What's in it



Classes are roughly balanced, so accuracy is not inflated by one dominant class.

Why it's clinically trustworthy

Gold-standard labels: each spectrum matched to histopathology.

Ex vivo: tissue sampled after removal at MUMC+, Maastricht.

Pseudonymised: patients appear only as ID numbers.

Open: published in Scientific Reports, CC BY 4.0.

Data: Vaysse et al., Scientific Reports 2020 · MUMC+, Maastricht

REIMS: reading the smoke

01 Burn
Diathermy vaporises a spot of tissue.

→

02 Aspirate
The smoke is sucked down a tube.

→

03 Ionise
Molecules in the smoke become charged ions.

→

04 Weigh
A time-of-flight mass spectrometer sorts the ions by mass.

→

05 Label
Histopathology confirms the tissue type.

One burn, one row

8,000 numbers: how much of each ion mass appeared, from m/z 200 to 1,000 in steps of 0.1. This range is dominated by lipids.

No extra step for the surgeon

The knife they already use makes the sample. Nothing is cut out, stained or sent away.

REIMS = rapid evaporative ionisation mass spectrometry, the "iKnife" principle

ARCHITECTURE

From smoke to a red or green light

THE MODEL · PCA-LDA-V2

Input
8,000 m/z channels

→

Scale
Standardise each channel

→

Compress
PCA, 100 components

→

Classify
Shrinkage LDA

→

Calibrate
Temperature scaling

→

Gate
Score ≥ 0.77, else Uncertain

THE DEMO SYSTEM

Smoke spectrum
Real held-out spectra, replayed on every burn

→

Laptop server
Flask runs the Python model, about 50 ms a call

→

iPad theatre
Apple Pencil is the diathermy tip; each cut shows the AI's call

→

Arduino LED
Red = tumour, green = fat, off = uncertain

Why a linear model? 689 samples against 8,000 features: classic chemometrics resists overfitting, can be explained, and folds into one 3 × 8,000 matrix small enough for embedded hardware.

Trained with scikit-learn · served by Flask · LED over USB serial

ACCURACY · TRAINING · SPLIT

Split by patient, never by spectrum



91.1%

nested cross-validation, folds grouped by patient

91.3%

held-out test: 146 of 160 spectra right, from patients it never saw

92.3%

published benchmark on the same data (leave-one-patient-out)

Why it matters: burns from one patient look alike. If a patient appeared in both training and test, accuracy would be inflated, so every split keeps each patient's burns together.

Benchmark: Vaysse et al., Scientific Reports 2020

FINDINGS AND RESULTS

Accurate, fast, and honest about doubt

91.3%

of tissue calls correct on patients the model had never seen

Fast enough for theatre

About 50 ms per spectrum in Python; about 0.1 ms once folded into one matrix.

Knows when it doesn’t know

Setting aside the least-confident 20% cut errors from 9.6% to 3.8%, and missed tumours from 13 to 1.

The hard cases

24 of 66 errors came with over 99% confidence. Boundaries, where tumour blends into stroma, are the weak spot.

Small enough to shrink

A model using just 40 key peaks kept about 90% accuracy: a path to a cheaper, targeted sensor.

Confidence and 40-peak analyses: leave-one-patient-out, all 689 spectra, baseline PCA-LDA

DATA SAFETY AND REGULATION

Safe by design, regulated as a device

Data safety

No identifiable data in the demo: public, pseudonymised spectra only.

Stays in the hospital: the model runs on a local machine, so nothing goes to the cloud.

UK GDPR: health data is special-category, so a DPIA and a direct-care lawful basis come first.

Traceable: every call carries the model version and its score.

Regulation

Medical device: software that guides surgery falls under UK MDR 2002 and the MHRA; we expect Class IIa or IIb.

Standards: ISO 13485 quality, IEC 62304 software lifecycle, ISO 14971 risk.

NHS: a DCB0129 clinical safety case and DTAC before any trust deploys it.

Surgeon in control: advisory only; an uncertain reading shows no colour.

Device class to be confirmed with the MHRA

BUSINESS MODEL

We sell fewer second operations

Who pays

NHS breast units and private hospitals. Breast surgeons champion it; trust procurement signs.

How we charge

A software licence per theatre per year [£__], plus a single-use smoke-sampling line per case [£__].

Who we partner with

Waters owns the REIMS technology behind the iKnife. We are the AI and theatre layer, not the instrument.

THE VALUE, PER 100 OPERATIONS

About 19 women need a second operation today. Halving that avoids about 9, worth roughly **£19,000** in direct costs alone.

THE LONG GAME

A 40-peak targeted sensor could replace the lab-grade spectrometer and cut the hardware bill.

Illustrative: 18.5% reoperation (Jeevan, BMJ 2012) × £2,136 extra each (Grant, BJS 2019)

CONCLUSION

The smoke already knows.

- Surgeons get the margin answer during the cut, not days later.
- 91.3% on unseen patients, and an honest “uncertain” when it isn’t sure.
- Next: live validation with a breast unit, sharper boundary detection, a 40-peak embedded sensor.

Thank you. Questions?